

Practice program

Q1 Student

10 marks

Write a program to enter the mark of a student in four subjects. Then calculate the total and aggregate. Display the grade obtained by the student, if student scores an aggregate greater than 75% then grade is distinction

Your Code

```
1 public class student{
2     public static void main(String[]args){
3         int m1,m2,m3,m4;
4         m1=80;
5         m2=75;
6         m3=90;
7         m4=85;
8         int total=m1+m2+m3+m4;
9         double aggregate=total/4.0;
10        System.out.println("mark 1= "+m1);
11        System.out.println("mark 2= "+m2);
12        System.out.println("mark 3= "+m3);
13        System.out.println("mark 4= "+m4);
14        System.out.println("total = "+total);
15        System.out.println("aggregate = "+aggregate+"%");
16        if (aggregate>75){
17            System.out.println("grade=distinction");
18        }
19        else{
20            System.out.println("grade = no distinction");
21        }
22    }
23 }
```

Run

Save

Input

stdin input

Output

```
mark 1= 80
mark 2= 75
mark 3= 90
mark 4= 85
total = 330
aggregate = 82.5%
```

Q2 ATM

10 marks

write a program to print the total amount available in the atm machine with the conditions applied total denominations are 2000, 500, 200,100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Your Code

```
1 public class ATM{
2     public static void main(String[]args){
3         int denomination =500;
4         int notes =20;
5         int total =denomination * notes;
6         System.out.println("denomination = "+denomination);
7         System.out.println("number of notes = "+notes);
8         System.out.println("total amount in ATM = Rs."+total);
9         if (denomination == 2000 || denomination == 500 ||denomination
10            System.out.println("valid denomination");
11        }
12        else{
13            System.out.println("invalid denomination");
14        }
15    }
16 }
```

Run

Save

Input

stdin input

Output

```
denomination = 500
number of notes = 20
total amount in ATM = Rs.10000
valid denomination
```

Q3 TAX

10 marks

Write a program to calculate tax given following condition :

- If income is less than or equal to 150000 then no tax
- If taxable income is 150001 – 300000 the charge 10% tax
- If taxable income is 300001 – 500000 the charge 20% tax
- If taxable income is above 500001 the charge 30% tax

Your Code

```
1 public class tax{
2     public static void main(String[]args){
3         double income =450000;
4         double tax;
5         if(income<=100000){
6             tax=0;
7         }
8         else if(income<=250000){
9             tax=income*0.10;
10        }
11        else if(income<=450000){
12            tax=income*0.20;
13        }
14        else{
15            tax=income*0.30;
16        }
17        System.out.println("income = Rs."+income);
18        System.out.println("tax =Rs."+tax);
19    }
20 }
```

**Input**

stdin input

Output

```
income = Rs.450000.0
tax =Rs.90000.0
```

Q4 BONUS

10 marks

a bonus 5% salary is given to grade A workers and 10% bonus to Grade B , write a program to enter salary and grade of employee , if salary less than employee gets an extra 2% bonus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get

Your Code

```
1 public class bonus{
2     public static void main(String[]args){
3         double salary=18000;
4         char grade='A';
5         double bonus=0;
6         if(grade=='A'){
7             bonus=salary*0.05;
8         }
9         else if(grade=='B'){
10            bonus=salary*0.10;
11        }
12        if(salary<20000){
13            bonus=bonus+(salary*0.02);
14        }
15        double totalsalary=salary+bonus;
16        System.out.println("salary=Rs. "+salary);
17        System.out.println("grade= "+grade);
18        System.out.println("bonus=Rs. "+bonus);
19        System.out.println("total salary=Rs. "+totalsalary);
20    }
21 }
```

**Input**

stdin input

Output

```
salary=Rs. 18000.0
grade= A
bonus=Rs. 1260.0
total salary=Rs. 19260.0
```